## Practical No.12

### Q.1] Apply Boundary Value Analysis (BVA) to test overload and floor limits.

Boundary value analysis for test overload

| Input Parameter | BVA Value (kg) |
|---|---|
| Below Maximum Weight | 100 |
| At Maximum Weight | 230 |
| Above Maximum Weight | 400 |

**Test cases:**

| Test Case ID | Objective | Steps (Input) | Expected Result | Actual Result | Status |
|---|---|---|---|---|---|
| TC_01 | Verify elevator operates safely below maximum load | Weight: 100 kg | Elevator moves smoothly without warning | Elevator moved normally | Pass |
| TC_02 | Verify elevator operates safely at maximum allowable load | Weight: 230 kg | Elevator moves with load indicator active | Elevator moved normally | Pass |
| TC_03 | Verify safety mechanism activates when load exceeds limit | Weight: 400 kg | Overload warning shown and elevator movement disabled | Error displayed | Pass |

## Boundary Value Analysis – Floor Limits

| Input Parameter | BVA Value |
|---|---|
| At minimum floor | 1 |
| At maximum floor | 15 |
| Below minimum floor | 0 |

## Test cases:

| Test Case ID | Objective | Steps (Input) | Expected Result | Actual Result | Status |
|---|---|---|---|---|---|
| TC_01 | Verify elevator operation at minimum floor boundary | Floor: 1 | Elevator moves to floor 1 without error | Elevator moved normally | Pass |
| TC_02 | Verify elevator operation at maximum floor boundary | Floor: 15 | Elevator moves to floor 15 without error | Elevator moved normally | Pass |
| TC_03 | Verify system blocks selection below minimum floor | Floor: 0 | Error message displayed: No such floor | Error displayed | Pass |